

Unit 16: Linearization and Differentials

1. The idea, in plain words

Here's a neat trick: if you zoom in close enough on any smooth curve, it starts to look just like a straight line — specifically, its tangent line. This unit is about using that tangent line as a stand-in for the actual curve, to make quick, easy estimates of a function's value near a point you already know well.

Picture it like this: you know $\sqrt{9} = 3$ exactly, no calculator needed. But what about $\sqrt{9.3}$? You could reach for a calculator — or you could notice that 9.3 is very close to 9, and use the tangent line to the curve $y = \sqrt{x}$ at the point $x = 9$ as a quick stand-in for the curve itself, right in that neighborhood. Since the tangent line is straight (and easy to compute), it gives you a fast, pretty accurate estimate without any heavy lifting.

This “stand-in tangent line” has a name: the linearization, $L(x)$. It's built from information you already know how to find: the function's value at some nearby “nice” point a , and the slope there ($f'(a)$).

A closely related idea: differentials. Instead of asking “what is f at this new point?”, differentials ask a slightly different but related question: “if x changes by a tiny amount, roughly how much does y change?” The answer, dy , is just the tangent line's rise for a given tiny run, dx . It's the same core idea as linearization, just phrased in terms of a small change rather than a specific new point.

2. Toolbox

Linearization (the tangent line at $x = a$, used as an approximation for $f(x)$ near a):

$$L(x) = f(a) + f'(a)(x - a)$$

Using it to estimate: for x close to a ,

$$f(x) \approx L(x) = f(a) + f'(a)(x - a)$$

Differentials: if dx represents a small change in x , the corresponding estimated change in y is

$$dy = f'(x) \cdot dx$$

The key distinction: Δy is the actual, exact change in y (which you'd get by plugging both x -values into f and subtracting). dy is the tangent line's estimate of that change. When dx is small, $dy \approx \Delta y$ — close, but not usually exactly equal.

Picking a good point a : always choose a nearby value where $f(a)$ and $f'(a)$ are both easy to compute by hand (a perfect square, a nice round number, an angle you know the trig values for, etc.).

3. Common mistakes

- Evaluating f' at the wrong point. Always evaluate the derivative at a (the known, nearby point) — not at the target value you're trying to estimate.

- Getting the sign of dx backward. If the target value is smaller than a , then dx is negative — don't forget the negative sign.
- Forgetting that $L(x)$ is only an estimate, not the exact answer. It'll be close to the true value for small dx , but it isn't meant to be perfectly exact — a bit of difference between the estimate and the true value is expected and normal.
- Choosing a bad reference point a . If $f(a)$ and $f'(a)$ are hard to compute, the whole point of the technique is lost — always pick the closest “nice” value.
- Forgetting the units when doing an error/estimation problem (like estimating volume error from a measurement error) — the differential dy carries the same units as y itself.

4. Problem Set

Warm-up

1. Let $f(x) = x^2$. Find the linearization $L(x)$ at $a = 3$, then use it to estimate $f(3.1)$.
2. Let $f(x) = x^3$. Find $L(x)$ at $a = 2$, then estimate $f(2.05)$.
3. Let $f(x) = \frac{1}{x}$. Find $L(x)$ at $a = 4$, then estimate $f(4.2)$.
4. Let $f(x) = \sqrt{x}$. Find $L(x)$ at $a = 9$, then estimate $f(9.3)$ (i.e., estimate $\sqrt{9.3}$).
5. Let $f(x) = x^2 + 3x$. Find $L(x)$ at $a = 1$, then estimate $f(1.1)$.
6. Let $f(x) = \sqrt{x}$. Find $L(x)$ at $a = 25$, then estimate $f(24.7)$ (i.e., estimate $\sqrt{24.7}$).

Standard

7. Let $y = x^3$. Find the differential dy when $x = 2$ and $dx = 0.1$.
8. Let $y = \sqrt{x}$. Find dy when $x = 16$ and $dx = -0.2$.
9. Estimate $\sqrt{50}$ using linearization at $a = 49$.
10. Estimate $\sqrt[3]{26}$ using linearization at $a = 27$ (i.e., $f(x) = x^{1/3}$).
11. Let $y = \frac{1}{x^2}$. Find dy when $x = 5$ and $dx = 0.05$.
12. Estimate $(3.98)^2$ using linearization at $a = 4$.

Challenge

13. A sphere's radius is measured as 10 cm, with a possible measurement error of ± 0.1 cm. Using differentials, estimate the resulting possible error in the computed volume, $V = \frac{4}{3}\pi r^3$.
14. A circle's radius is measured as 5 cm, with a possible error of ± 0.05 cm. Using differentials, estimate the resulting error in the computed area, $A = \pi r^2$, and express it as an approximate percentage error.
15. Let $f(x) = \tan x$. Find $L(x)$ at $a = \frac{\pi}{4}$, then use it to estimate $f\left(\frac{\pi}{4} + 0.01\right)$.
16. Let $f(x) = \sin x$. Find $L(x)$ at $a = 0$, then use it to estimate $\sin(0.05)$.
17. A cube's side length is measured as 6 cm, with a possible error of ± 0.05 cm. Using differentials, estimate the resulting error in the computed volume, $V = s^3$, and express it as an approximate percentage error.

Applied

18. A company's cost (in dollars) to produce x units is $C(x) = 0.01x^2 + 5x + 200$. Use a differential to estimate the change in cost when production increases from 100 to 101 units. Then compute

the exact change, $C(101) - C(100)$, and compare.

19. A bacteria population is modeled by $P(t) = 500 + 20t + t^2$. Use a differential to estimate the change in population from $t = 10$ to $t = 10.2$.
20. A rocket's height (in feet) is modeled by $h(t) = -16t^2 + 200t + 10$. Use a differential to estimate the change in height from $t = 3$ to $t = 3.1$.
21. A company's revenue (in dollars) is $R(x) = 100x - 0.5x^2$. Use a differential to estimate the change in revenue when sales increase from $x = 50$ to $x = 51$. Then compute the exact change, $R(51) - R(50)$, and compare.